

How Many Steps to Get Numbers?

Three Paths — Pick the One That Fits Your Workflow

Option 1: Zero Steps

Every example ships with a pre-generated

`METRICS_TABLE.md`

`examples/yolov12/METRICS_TABLE.md`

- CPU, GPU, NPU throughput & latency
- Cross-device comparison table
- Auto-generated from benchmark runs
- **Plan:** expose as a web dashboard

Just open the file — numbers are already there

Option 2: Docker (One Command)

Pre-installed ROCm + RyzenAI environment

```
# Pull & run the container
docker run physical-ai-sdk

# Inside the container:
make benchmark-all-devices metric
```

- No local installs needed
- Reproducible environment
- Generates `METRICS_TABLE.md` automatically

One command to fresh numbers

Option 3: Manual Setup

Full local installation — 4 steps:

```
# 1. Install ROCm
./install_rocm_stack.sh

# 2. Install RyzenAI
./install_ryzen_ai_source_stack.sh

# 3. Navigate to example
cd examples/yolov12

# 4. Run benchmarks
make benchmark-all-devices metric
```

Output: `METRICS_TABLE.md`

Most control, most setup

Easy to Use

Uniform Interface Across Every Example

Make-Based Uniformity

Same commands work in **every** example directory:

```
make benchmark-all-devices metric # all devices
make benchmark-gpu                # GPU only
make benchmark-cpu                # CPU only
make benchmark-npu                # NPU only
make eval                          # evaluation pipeline
```

- Learn once, use everywhere
- No per-model setup surprises
- Consistent output format (`METRICS_TABLE.md`)

Every Example Includes:

Feature	Description
CPU / GPU / NPU flows	Benchmark each compute device for throughput & latency
Eval pipeline	Sample inputs → sample outputs to visually verify correctness
Guiding documentation	Per-example tutorial to walk you from setup to results

Principle: If you can run one example, you can run them all — same commands, same structure, same outputs.

Pain Points

Current Challenges on Edge Devices

ROCm Installation Stability

- ROCm install on edge devices is **not yet stable**
- Driver/firmware mismatches on some hardware revisions
- Workaround: pinned ROCm versions in install scripts

Reboot During Installation

- ROCm installation requires a **system reboot** mid-process
- Breaks unattended / CI-driven installs
- Users unfamiliar with the flow find this unexpected

Sudo / Root Access Required

- Both ROCm and RyzenAI installers need **sudo privileges**
- Blocks usage in locked-down enterprise environments
- Containers mitigate this but add their own complexity

Pain Point	Impact	Mitigation
Unstable ROCm install	Blocked users	Pinned versions
Mid-install reboot	CI breakage	Docker path
Sudo required	Enterprise friction	Container deploy

Docker path sidesteps all three issues — pre-built image with ROCm + RyzenAI already configured, no reboot, no sudo on host.